B6.1 Monitoring and maintaining the environment

Summary

Living organisms interact with each other, the environment and with humans in many different ways. If the variety of life is to be maintained we must actively manage our interactions with the environment. We must monitor our environment, collecting and interpreting information about the natural world, to identify patterns and relate possible cause and effect.

Underlying knowledge and understanding

From their study in topic B4, learners should be familiar with ecosystems and the various ways organisms interact. They should understand how biotic and abiotic factors influence communities. Learners should be familiar with the gases of the atmosphere from Key Stage 3.

Common misconceptions

It is important that in the study of this topic learners are given opportunities to explore both positive and negative human interactions within ecosystems.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM6.1i	calculate arithmetic means	M2b
BM6.1ii	plot and draw appropriate graphs selecting appropriate scales for the axes	M4a and M4c
BM6.1iii	understand and use percentiles	M1c
BM6.1iv	extract and interpret information from charts, graphs and tables	M2c and M4a
BM6.1v	understand the principles of sampling as applied to scientific data	M2d

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
B6.1a explain how to carry out a field investigation into the distribution and abundance of organisms in a habitat and how to determine their numbers in a given area	sampling techniques (random and transects, capture-recapture), use of quadrats, pooters, nets, keys and scaling up methods	M2c, M2d, M3a	WS1.2d, WS1.2b, WS1.2c, WS1.2e, WS1.3h, WS2a, WS2b, WS2c, WS2d	Investigation of ecological sampling methods. Using the symbols =, $<$, $<<$, $>>$, $>$, $\propto$, ~ in answers where appropriate. (PAG B1, PAG B3) Investigation of sampling using a suitable model (e.g. measuring the red sweets in a mixed selection).
B6.1b describe both positive and negative human interactions within ecosystems and explain their impact on biodiversity	the conservation of individual species and selected habitats and threats from land use and hunting		WS2a, WS2b, WS2c, WS2d	Investigation into the effectiveness of germination in different strengths of acid rain. (PAG B3, PAG B6) Investigation into the effects of lichen distribution against pollution. (PAG B3)
B6.1c explain some of the benefits and challenges of maintaining local and global biodiversity	the difficulty in gaining agreements for and the monitoring of conservation schemes along with the benefits of ecotourism			
B6.1d ☑ **evaluate the evidence for the impact of environmental changes on the distribution of organisms, with reference to water and atmospheric gases**				